

Chapter 6

Predictive Analytics II: Text, Web, and Social Media Analytics

Learning Objectives (1 of 2)

- 6.1** Describe **text mining** and understand the need for text mining
- 6.2** Differentiate among **text analytics**, text mining, and data mining
- 6.3** Understand the different **application areas** for text mining
- 6.4** Know the **process** of carrying out a text mining project
- 6.5** Appreciate the **different methods** to introduce structure to text-based data

Learning Objectives (2 of 2)

6.6 Describe **sentiment analysis**

6.7 Develop familiarity with popular **applications of sentiment analysis**

6.8 Learn the **common methods** for sentiment analysis

6.9 Become familiar with **speech analytics** as it relates to sentiment analysis

Text Analytics VS Text Mining

Text Analytics and Text Mining (1 of 2)

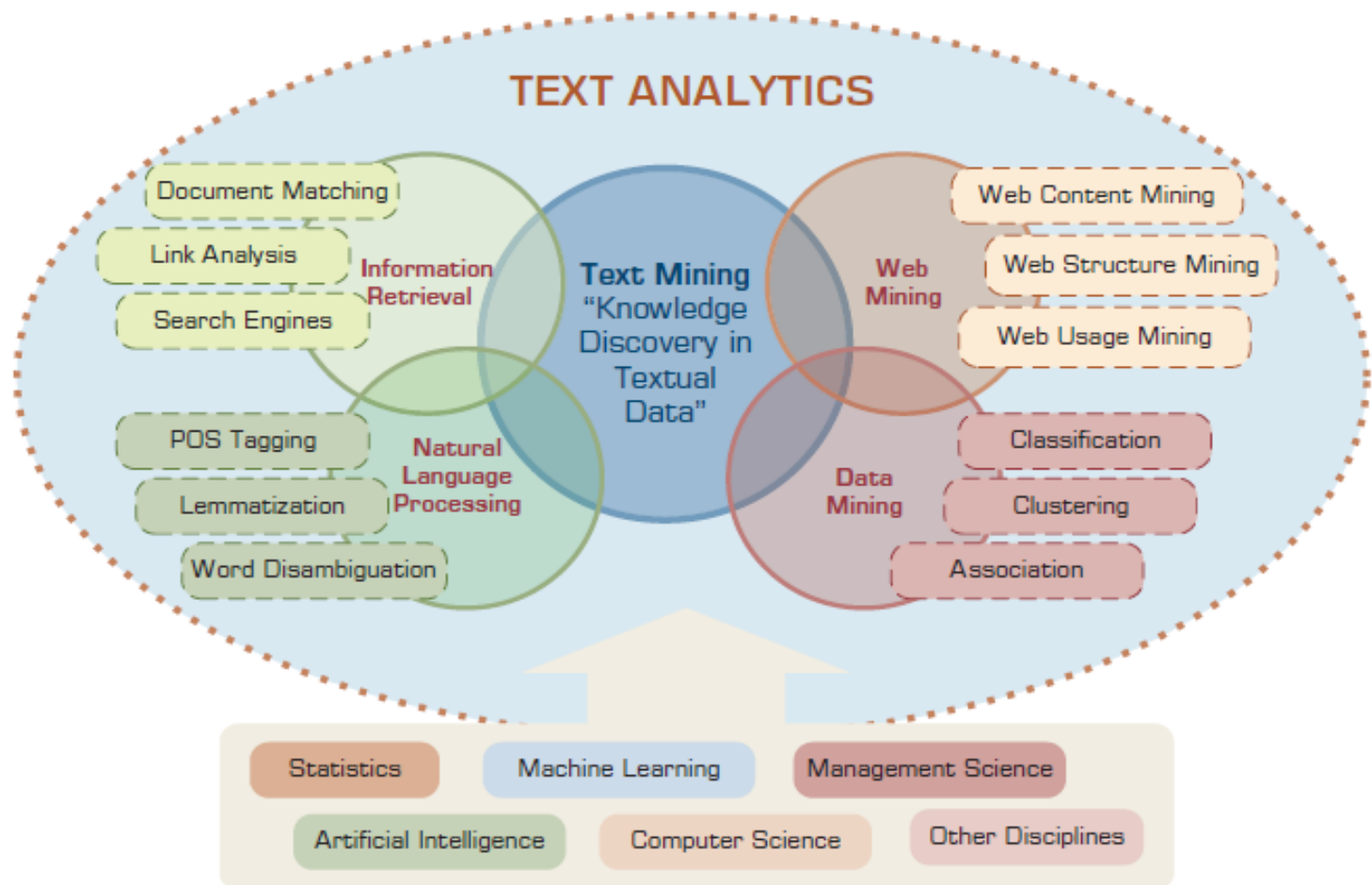
- Text Analytics versus Text Mining
- Text Analytics =
 - Information Retrieval +
 - Information Extraction +
 - Data Mining +
 - Web Mining

or simply

Text Analytics = Information Retrieval + Text Mining

Text Analytics and Text Mining (2 of 2)

- **Figure 6.2** Text Analytics, Related Application Areas, and Enabling Disciplines



Data Mining vs Text Mining

Data Mining Versus Text Mining

- Both seek for novel and useful patterns
- Both are semi-automated processes
- Difference is the nature of the data:
 - Structured versus unstructured data
 - **Structured data:** in databases
 - **Unstructured data:** Word documents, PDF files, text excerpts, XML files, and so on
- **To perform text mining** – first, impose structure to the data, then mine the structured data

Text Mining Concepts (1 of 2)

- 85-90 percent of all corporate data is in some kind of unstructured form (e.g., text)
- *Unstructured corporate data* is doubling in size every 18 months
- Tapping into these information sources is not an option, but a need to stay competitive
- Answer: text mining
 - A semi-automated process of extracting knowledge from unstructured data sources
 - a.k.a. text data mining or knowledge discovery in textual databases

Text Mining Concepts (2 of 2)

- Benefits of text mining are obvious especially in **text-rich data environments**
 - e.g., law (court orders), academic research (research articles), finance (quarterly reports), medicine (discharge summaries), biology (molecular interactions), technology (patent files), marketing (customer comments), etc.
- Electronic communication records (e.g., e-mail)
 - Spam filtering
 - E-mail prioritization and categorization
 - Automatic response generation

Text Mining Application Area

- Information extraction
- Topic tracking
- Summarization
- Categorization
- Clustering
- Concept linking
- Question answering

Text Mining Terminology (1 of 2)

- Unstructured or semistructured data
- Corpus (and corpora)
- Terms
- Concepts
- Stemming
- Stop words (and include words)
- Synonyms (and polysemes)
- Tokenizing

Text Mining Terminology (2 of 2)

- Term dictionary
- Word frequency
- Part-of-speech tagging
- Morphology
 - The study of internal structure of words
- Term-by-document matrix
 - Occurrence matrix
- Singular value decomposition
 - Latent semantic indexing

Application Case 6.1

Insurance Group Strengthens Risk Management with Text Mining Solution

Questions for Discussion

1. How can text analytics and mining be used to keep up with changing business needs of insurance companies?
 - bodily injury claims
2. What were the challenges, the proposed solution, and the obtained results?
 - Dell Statistica Data Miner & Statistica Text Miner
3. Can you think of other uses of text analytics and text mining for insurance companies?

Natural Language Processing (NLP) (1 of 4)

- Structuring a collection of text
 - **Old approach:** bag-of-words
 - **New approach:** natural language processing
- NLP is ...
 - a very important concept in text mining
 - a subfield of artificial intelligence and computational linguistics
 - the studies of "understanding" the natural human language
- Syntax versus semantics-based text mining

Natural Language Processing (NLP) (2 of 4)

- What is “Understanding”?
 - Human understands, what about computers?
 - Natural language is vague, context driven
 - True understanding requires extensive knowledge of a topic
 - **Can/will computers ever understand natural language the same/accurate way we do?**

Natural Language Processing (NLP) (3 of 4)

- **Challenges** in NLP
 - Part-of-speech tagging
 - Text segmentation
 - Word sense disambiguation
 - Syntax ambiguity
 - Imperfect or irregular input
 - Speech acts
- Dream of AI community
 - to have algorithms that are capable of automatically reading and obtaining knowledge from text

Natural Language Processing (NLP) (4 of 4)

- **WordNet**

- A laboriously hand-coded database of English words, their definitions, sets of synonyms, and various semantic relations between synonym sets
- A major resource for NLP
- Need automation to be completed

- **Sentiment Analysis**

- A technique used to detect favorable and unfavorable opinions toward specific products and services
- SentiWordNet

Application Case 6.2 (1 of 2)

AMC Networks Is Using Analytics to Capture New Viewers, Predict Ratings, and Add Value for Advertisers in a Multichannel World

A Web-Based Dashboard Used by AMC Networks



[Source: AMC Networks]

Application Case 6.2 (2 of 2)

Questions for Discussion

1. What are the common challenges broadcasting companies are facing nowadays? How can analytics help to alleviate these challenges?
2. How did AMC leverage analytics to enhance their business performance?
3. What were the types of text analytics and text mining solutions developed by AMC networks? Can you think of other potential uses of text mining applications in the broadcasting industry?

NLP Task Categories

- Question answering
- Automatic summarization
- Natural language generation & understanding
- Machine translation
- Foreign language reading & writing
- Speech recognition
- Text proofing, optical character recognition
- Optical character recognition

Text Mining Applications

- Marketing applications
 - Enables better CRM
- Security applications
 - ECHELON, OASIS
 - Deception detection (...)
- Medicine and biology
 - Literature-based gene identification (...)
- Academic applications
 - Research stream analysis

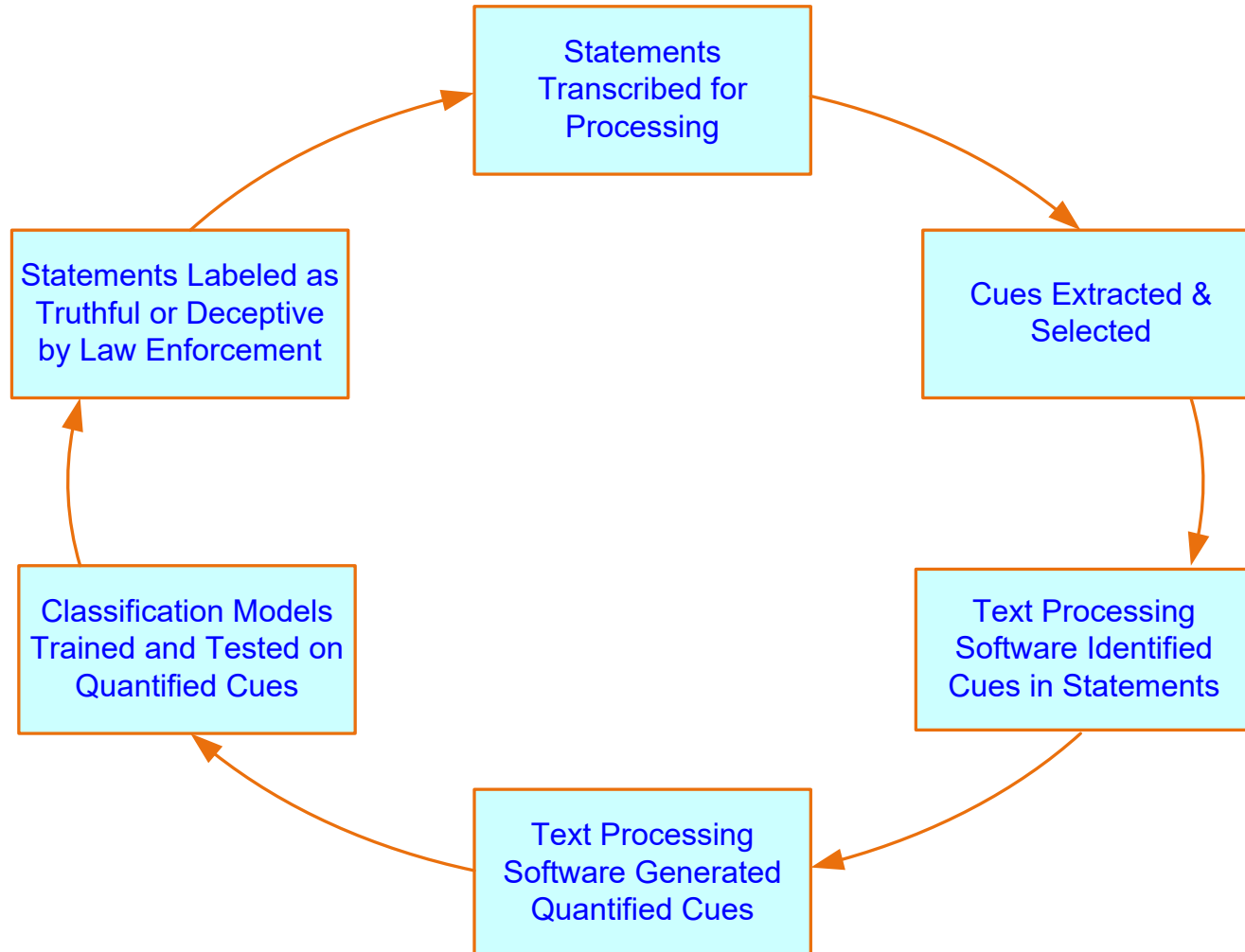
Application Case 6.3 (1 of 4)

Mining for Lies

- Deception detection
 - A difficult problem
 - If detection is limited to only text, then the problem is even more difficult
- The study
 - analyzed text-based testimonies of person of interests involved in crimes at military bases
 - used only text-based features (cues)

Application Case 5.3 (2 of 4)

- Figure 5.3 Text-Based Deception-Detection Process**



Application Case 6.3 (3 of 4)

Table 5.1 Categories and Examples of Linguistic Features Used in Deception Detection

Category	Example Cues
Quantity	Verb count, noun-phrase count, etc.
Complexity	Average number of clauses, average sentence length, etc.
Uncertainty	Modifiers, modal verbs, etc.
Nonimmediacy	Passive voice, objectification, etc.
Expressivity	Emotiveness
Diversity	Lexical diversity, redundancy, etc.
Informality	Typographical error ratio
Specificity	Spatiotemporal information, perceptual information, etc.
Affect	Positive affect, negative affect, etc.

Application Case 6.3 (4 of 4)

- 371 usable statements are generated
- 31 features are used
- Different feature selection methods used
- 10-fold cross validation is used
- Results (overall % **accuracy**)

Logistic regression 67.28

Decision trees 71.60

Neural networks 73.46

Application Case 5.4

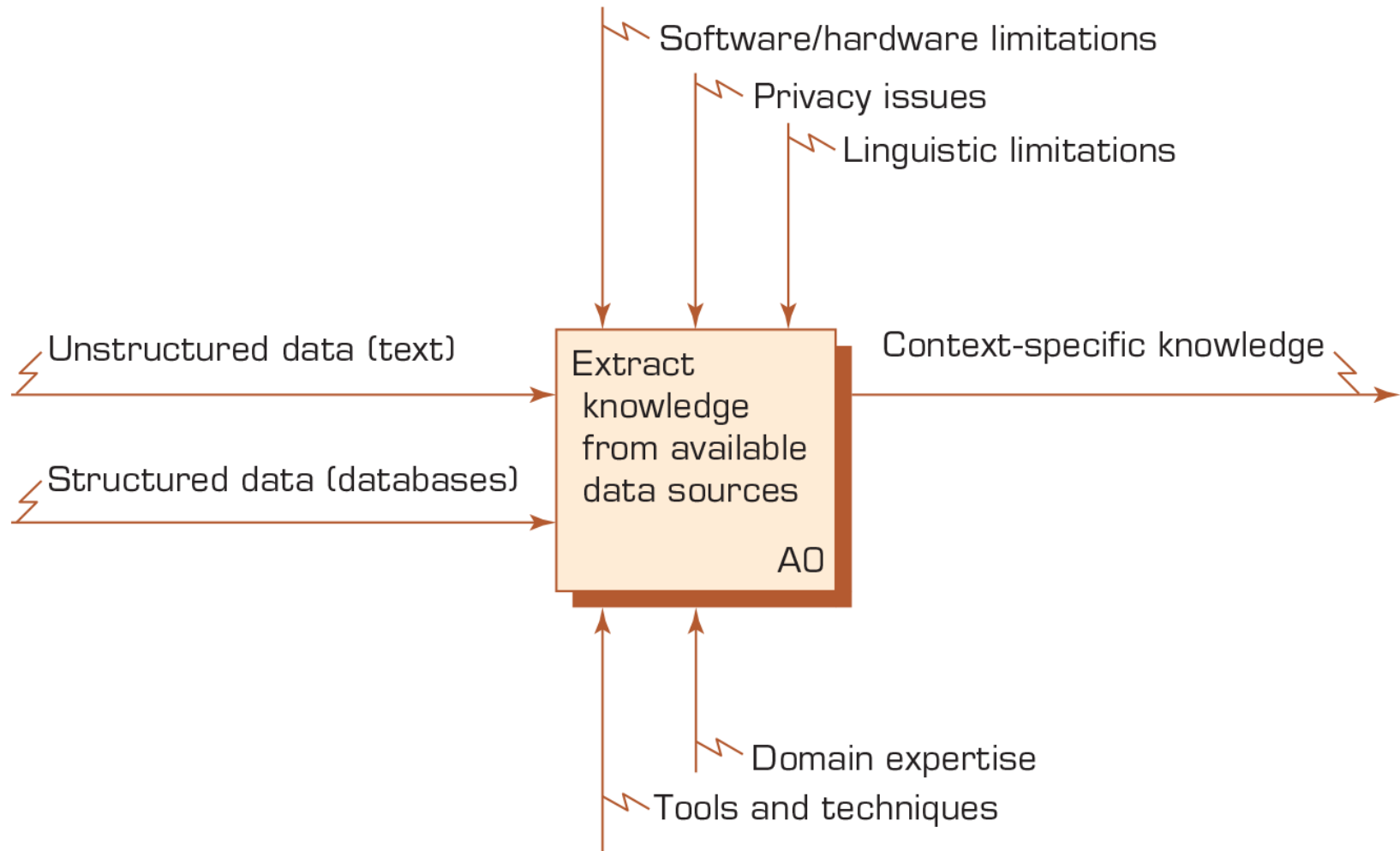
Bringing the Customer into the Quality Equation: Lenovo Uses Analytics to Rethink Its Redesign

Questions for Discussion

1. How did Lenovo use *text analytics and text mining* to **improve quality and design of their products** and ultimately improve customer satisfaction?
 - customer sentiment, warranty, call center analysis
1. What were the challenges, the proposed solution, and the obtained results?

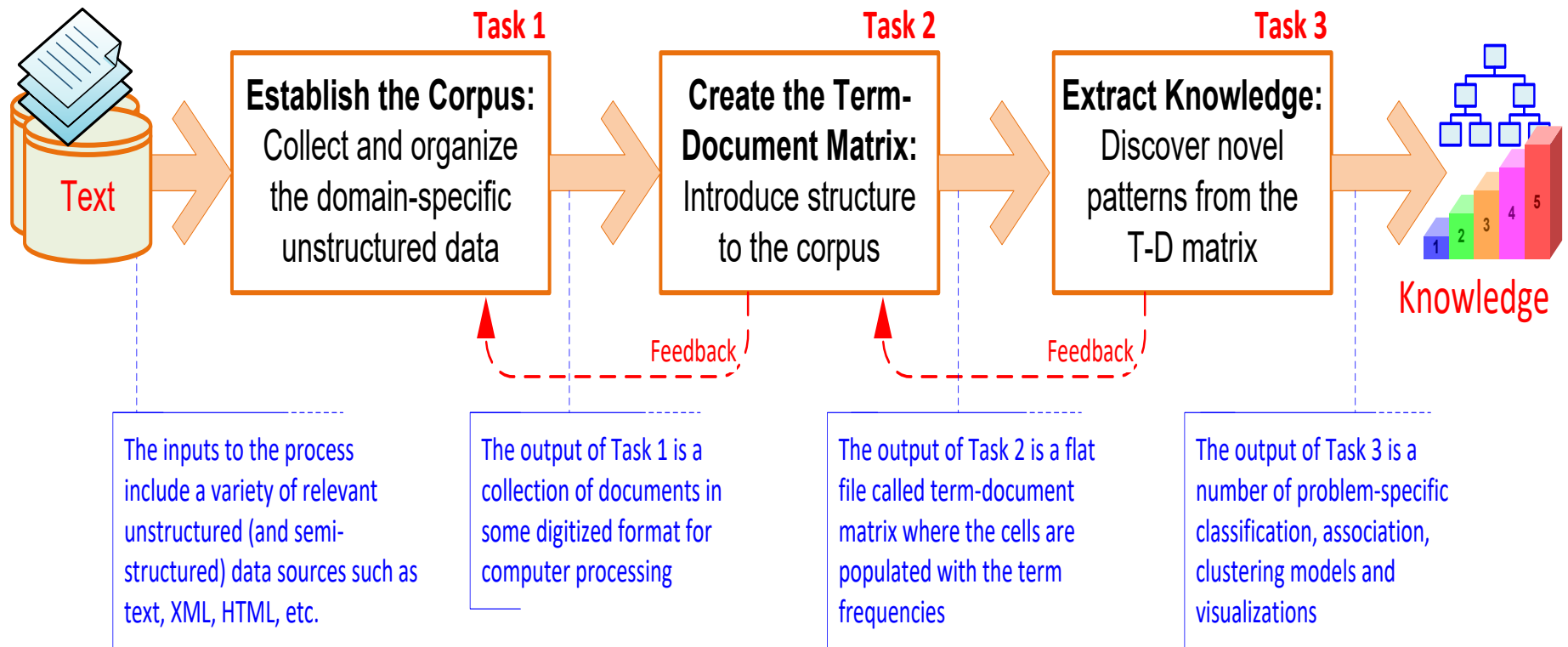
Text Mining Process (1 of 7)

- A Context Diagram for **Text Mining Process**



Text Mining Process (2 of 7)

- **Figure 5.6** The Three-Step/Task Text Mining Process



Text Mining Process (3 of 7)

- **Step 1:** Establish the corpus
 - Collect all relevant unstructured data (e.g., textual documents, XML files, e-mails, Web pages, short notes, voice recordings...)
 - Digitize, standardize the collection (e.g., all in ASCII text files)
 - Place the collection in a common place (e.g., in a flat file, or in a directory as separate files)

Text Mining Process (4 of 7)

- **Step 2:** Create the Term-by-Documents Matrix (TDM)

Terms Documents	investment risk	project management	software engineering	development	SAP	...
Document 1	1			1		
Document 2		1				
Document 3			3		1	
Document 4		1				
Document 5			2	1		
Document 6	1			1		
...						

Text Mining Process (5 of 7)

- Should all terms be included?
 - Stop words, include words
 - Synonyms, **homonyms**
 - Stemming
- What is the best representation of the indices (values in cells)?
 - Row counts; binary frequencies; log frequencies;
 - Inverse document frequency

Text Mining Process (6 of 7)

- **TDM** is a sparse matrix. How can we reduce the dimensionality of the TDM?
 - Manual - a domain expert goes through it
 - Eliminate terms with very few occurrences in very few documents (?)
 - Transform the matrix using singular value decomposition (SVD)
 - SVD is similar to principle component analysis

Text Mining Process (7 of 7)

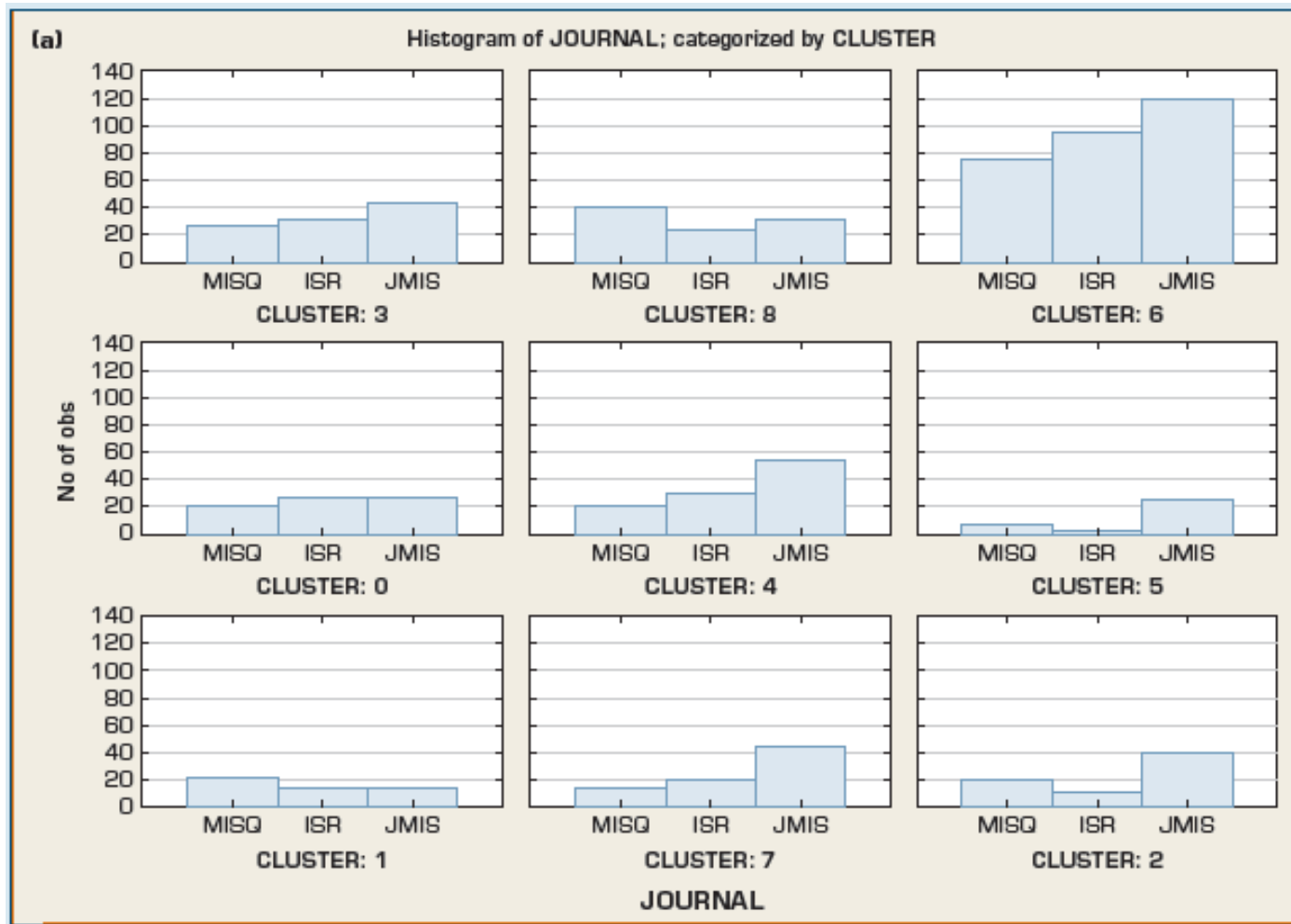
- **Step 3:** Extract patterns/knowledge
 - Classification (text categorization)
 - Clustering (natural groupings of text)
 - Improve search recall
 - Improve search precision
 - Scatter/gather
 - Query-specific clustering
 - Association
 - Trend Analysis (...)

Application Case 6.5 (1 of 2)

Research Literature Survey with Text Mining

- Mining the published IS literature
 - MIS Quarterly (MISQ)
 - Journal of MIS (JMIS)
 - Information Systems Research (ISR)
 - Covers 12-year period
 - 901 papers are included in the study
 - Only the paper **abstracts** are used
 - **9 clusters** are generated for further analysis

Application Case 6.5 (2 of 2)



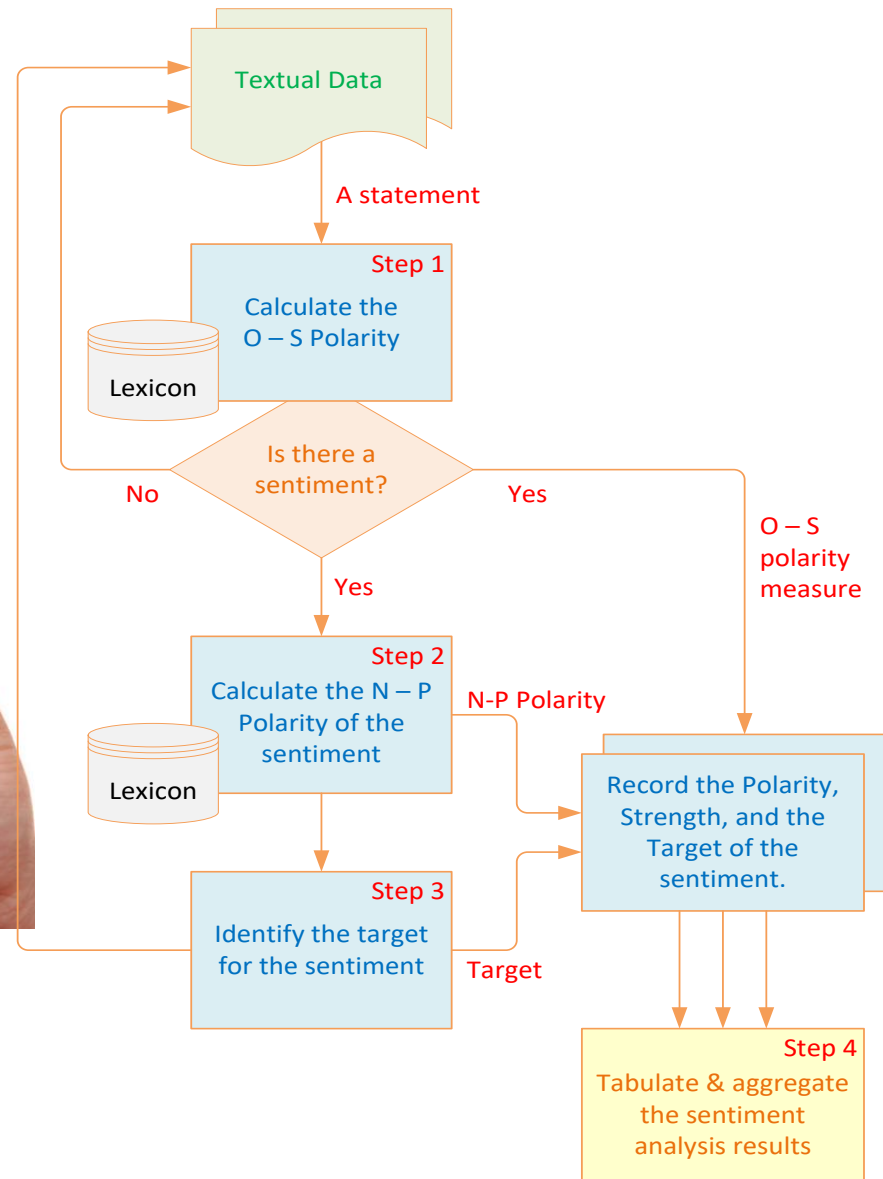
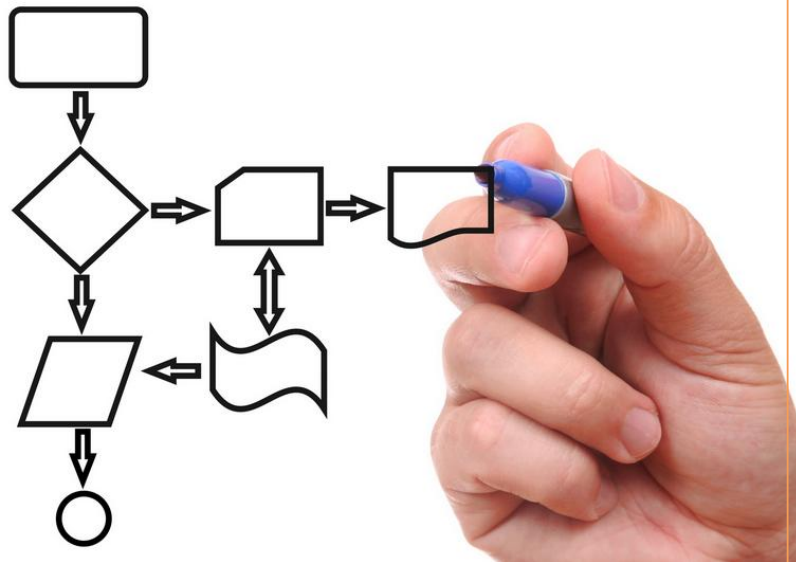
Sentiment Analysis

- Sentiment → belief, view, opinion, and conviction
- Sentiment analysis is trying to **answer** the question “What do people feel about a certain topic?”
- By analyzing data related to opinions of many using a variety of automated tools
- Used in variety of domains, but its applications in CRM are especially noteworthy (which related to customers/consumers’ opinions)

Sentiment Analysis Applications

- Voice of the customer (VOC)
- Voice of the Market (VOM)
- Voice of the Employee (VOE)
- Brand Management
- Financial Markets
- Politics
- Government Intelligence
- ... others

Sentiment Analysis Process (1 of 3)



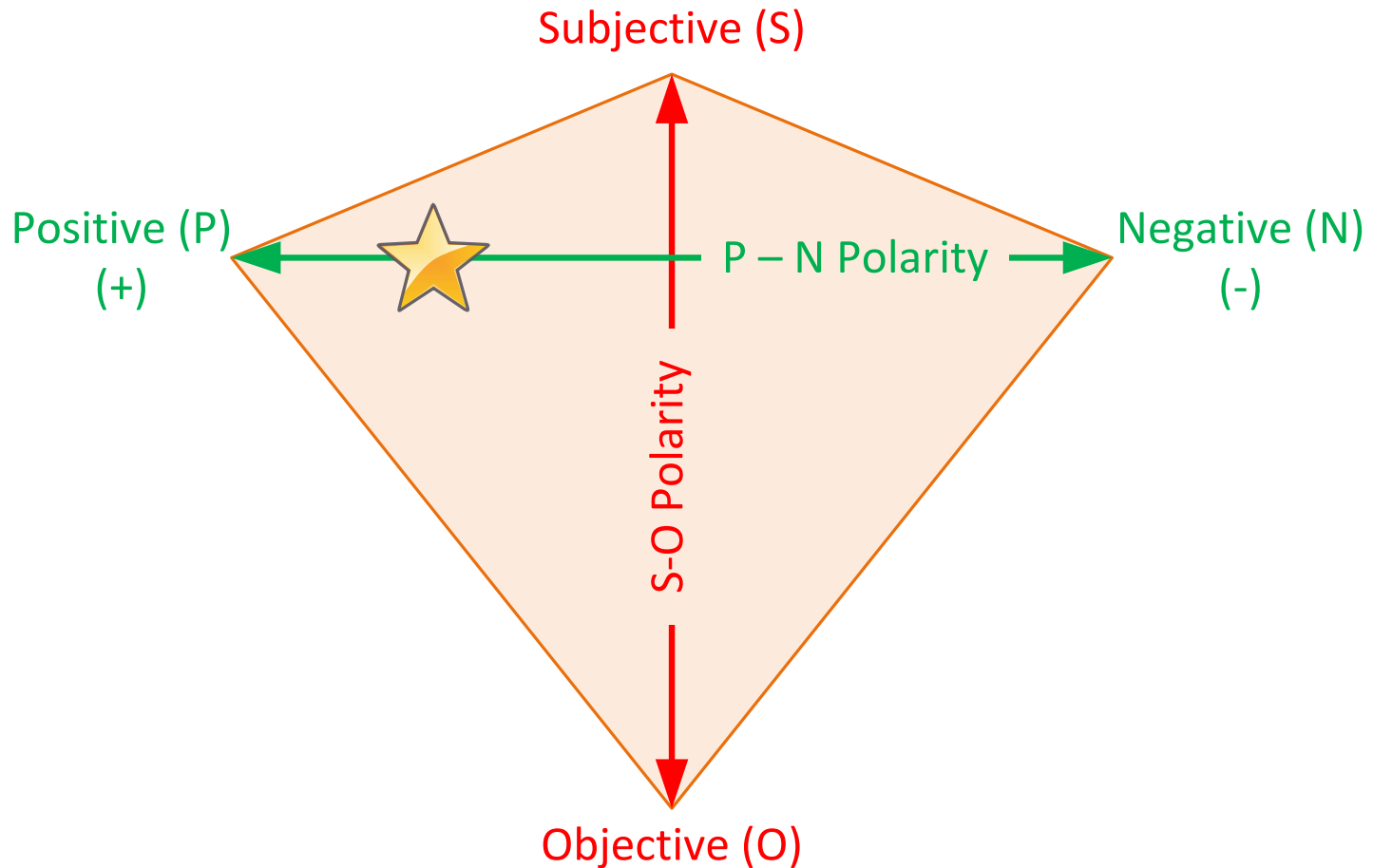
Sentiment Analysis Process (2 of 3)

- **Step 1 – Sentiment Detection**
 - Comes right after the retrieval and preparation of the text documents
 - It is also called detection of objectivity
 - **Fact [= objectivity] versus Opinion [= subjectivity]**
- **Step 2 – N-P Polarity Classification**
 - Given an opinionated piece of text, the goal is to classify the opinion as falling under one of two opposing sentiment polarities
 - **N [= negative] versus P [= positive]**

Sentiment Analysis Process (3 of 3)

- **Step 3 – Target Identification**
 - The goal of this step is to accurately identify the target of the expressed sentiment (e.g., a person, a product, an event, etc.)
 - Level of difficulty → the application domain
- **Step 4 – Collection and Aggregation**
 - Once the sentiments of all text data points in the document are identified and calculated, they are to be aggregated
 - Word → Statement → Paragraph → Document

P-N Polarity and S-O Polarity



Topic Modeling (1 of 3)

- Also known as “topic detection”
- Discover the topics (i.e., themes) in a give corpus
- Employs and unsupervised discovery process
- Often uses probabilistic ML algorithms
- General assumptions:
 1. each document consists of a mix of topics, and
 2. each topic consists of a collection of words/terms; the topics are “hidden” or “latent” constructs in between documents and words

Topic Modeling (2 of 3)

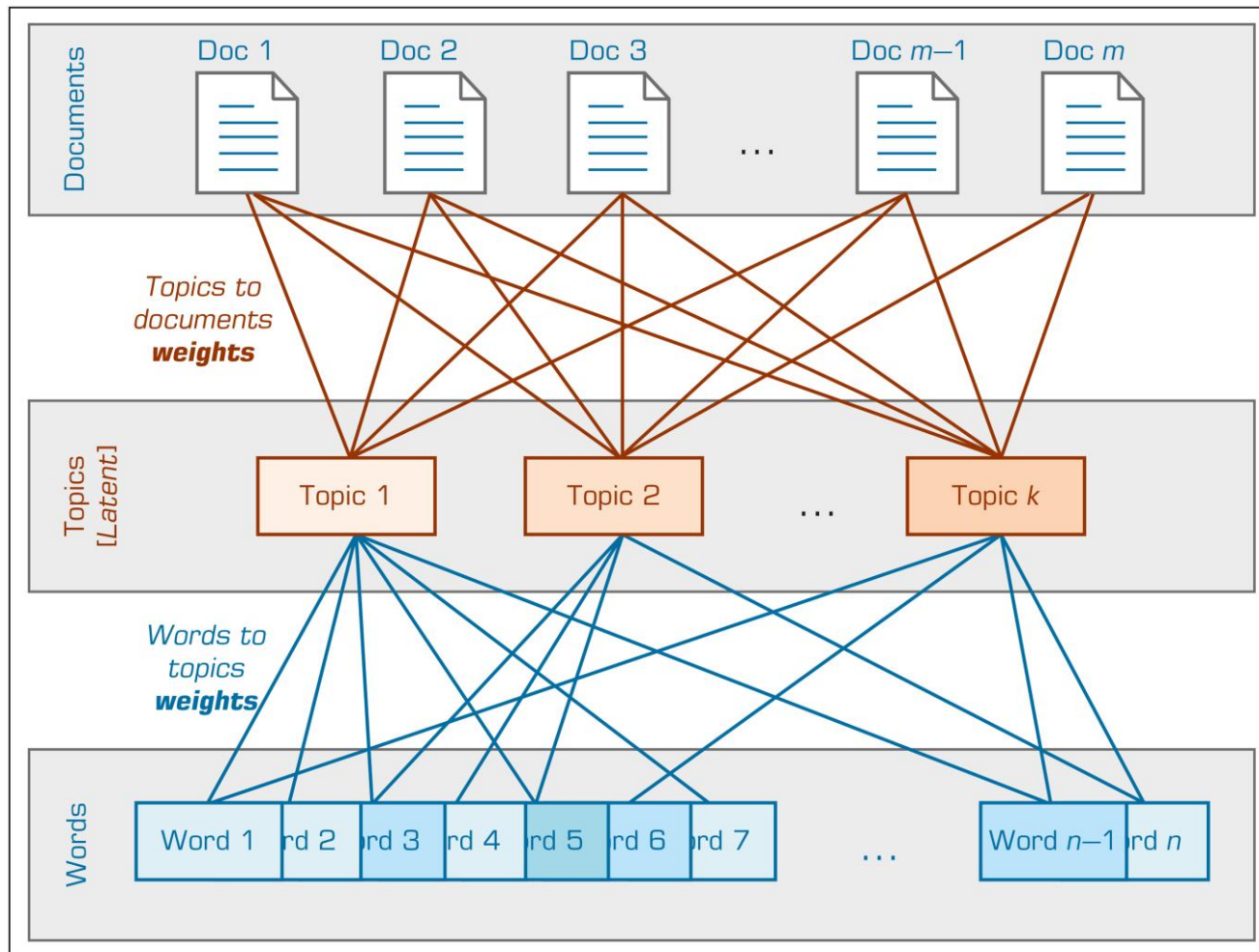
- Several methods used in topic modeling
 1. Simple clustering based on frequency of words in documents using term-document matrix
 2. Latent Semantic Indexing/Analysis (LSI/LSA) using **TF-IDF** and Singular Value Decomposition (SVD)

$$\underbrace{[tf - idf]_{i,j}}_{\text{Score for document } i \text{ and term } j} = \underbrace{[tf]_{i,j}}_{\substack{\text{Number of} \\ \text{occurrences} \\ \text{of term } j \text{ in} \\ \text{document } i}} \times \log \frac{\underbrace{N}_{\substack{\text{Total number} \\ \text{of documents}}}}{\underbrace{[df]_j}_{\substack{\text{Number of} \\ \text{document} \\ \text{containing} \\ \text{the term } j}}}$$

3. Latent Dirichlet Allocation (LDA)

Topic Modeling (3 of 3)

- LDA is among the most popular topic modeling techniques

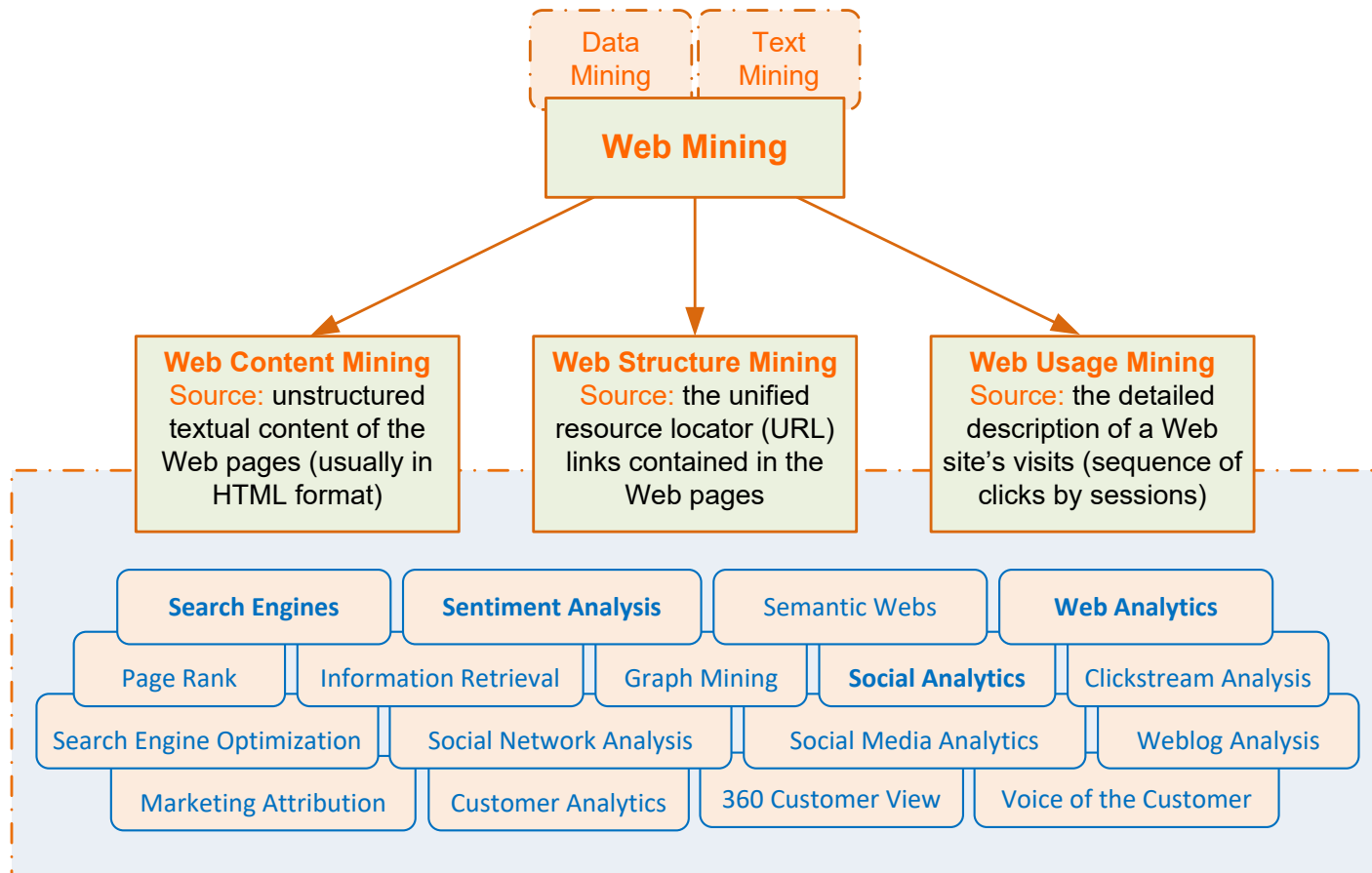


Web Mining Overview

- Web is the largest repository of data
- Data is in HTML, XML, text format
- Challenges (of processing Web data)
 - The Web is too big for effective data mining
 - The Web is too complex
 - The Web is too dynamic
 - The Web is not specific to a domain
 - The Web has everything
- **Opportunities and challenges are great!**

Web Mining

Web mining (or Web data mining) is the **process** of discovering intrinsic relationships from Web data (textual, linkage, or usage)



Web Content/Structure Mining

- Mining the textual content on the Web
- Data collection via Web crawlers
- Web pages include hyperlinks
 - Authoritative pages
 - Hubs
 - Hyperlink-induced topic search (HITS) alg.

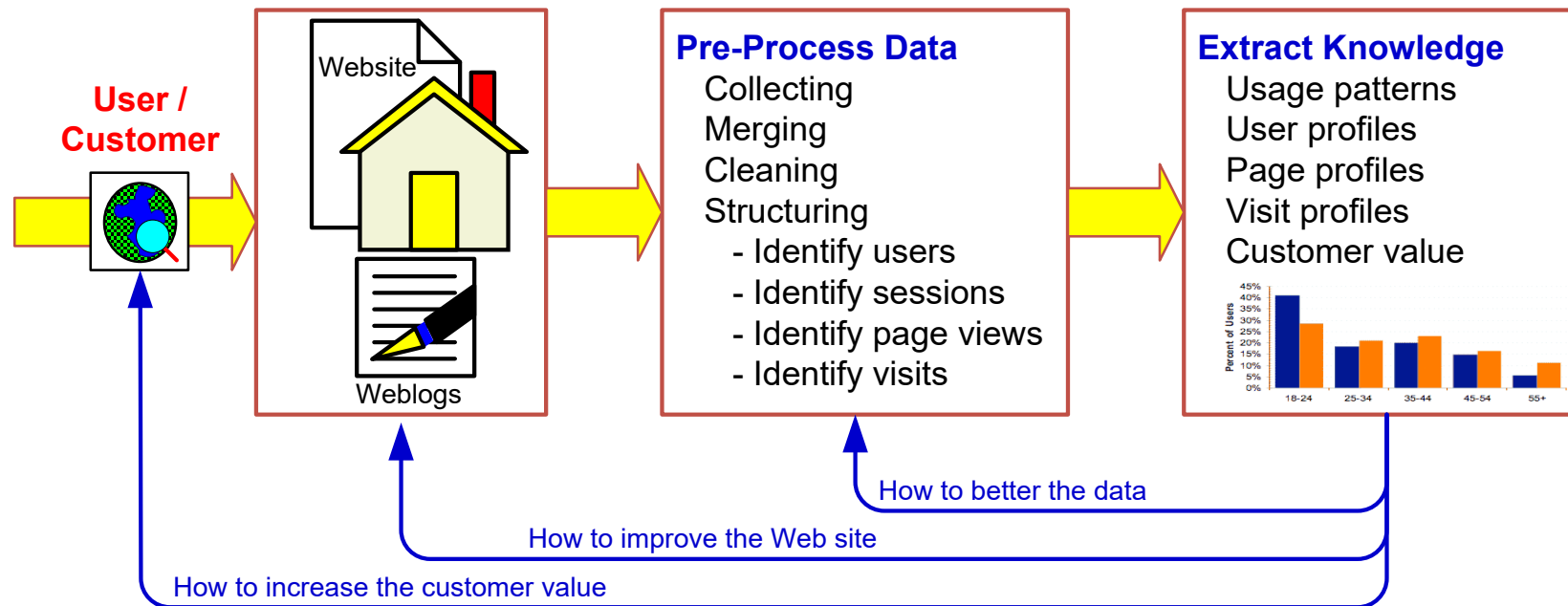
Web Usage Mining (1 of 2)

- Extraction of information from data generated through Web page visits and transactions...
 - data stored in server access logs, referrer logs, agent logs, and client-side cookies
 - user characteristics and usage profiles
 - metadata, such as page attributes, content attributes, and usage data
- Clickstream data
- Clickstream analysis

Web Usage Mining (2 of 2)

- Web usage mining **applications**
 - Determine the lifetime value of clients
 - Design cross-marketing strategies across products.
 - Evaluate promotional campaigns
 - Target electronic ads and coupons at user groups based on user access patterns
 - Predict user behavior based on previously learned rules and users' profiles
 - Present dynamic information to users based on their interests and profiles
 - ...

Web Usage Mining (Clickstream Analysis)



Web Analytics Metrics (1 of 3)

- **Web site usability**
 - How were the visitors using my Web site?
- **Traffic sources**
 - Where did they come from?
- **Visitor profiles**
 - What do my visitors look like?
- **Conversion statistics**
 - What does it all mean for the business?

Web Analytics Metrics (2 of 3)

Web Site Usability

- Page views
- Time on site
- Downloads
- Click map
- Click paths

Traffic Source

- Referral Web sites
- Search engines
- Direct
- Offline campaigns
- Online campaigns

Web Analytics Metrics (3 of 3)

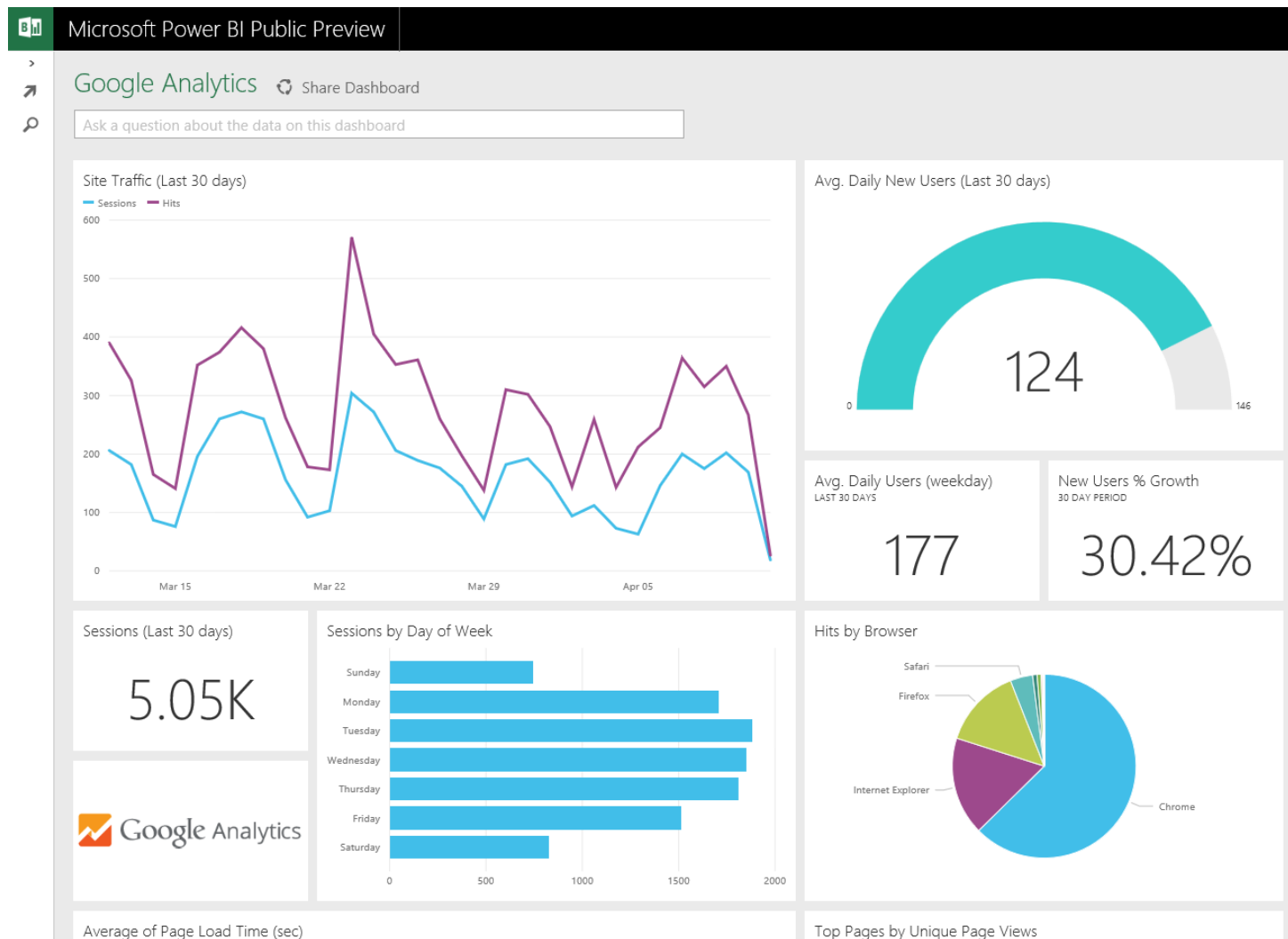
Visitor Profiles

- Keywords
- Content groupings
- Geography
- Time of day
- Landing page profiles

Conversion Statistics

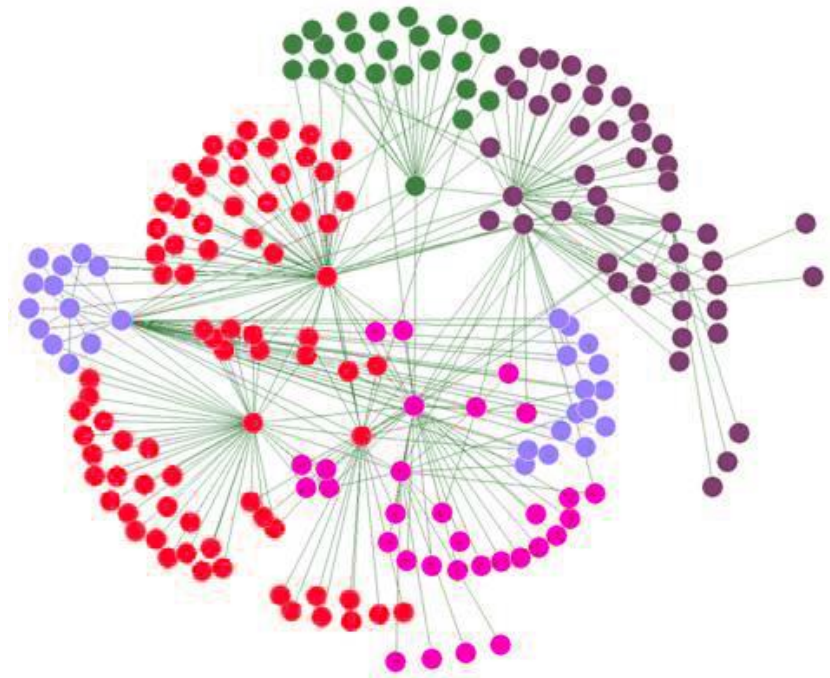
- New visitors
- Returning visitors
- Leads
- Sales/conversions
- Abandonment/exit rate

A Sample Web Analytics Dashboard



Social Analytics /Social Network Analysis (1 of 2)

- **Social Network** - social structure composed of individuals linking to each other
- Analysis of social dynamics
- Interdisciplinary field
 - Social psychology
 - Sociology
 - Statistics
 - Graph theory



Social Analytics /Social Network Analysis (2 of 2)

- **Social Networks help study** relationships between individuals, groups, organizations, societies
 - **Self organizing**
 - **Emergent**
 - **Complex**
- Typical social network types
 - Communication networks, community networks, criminal networks, innovation networks, ...

Application Case 6.8

Tito's Vodka Establishes Brand Loyalty with an Authentic Social Strategy

Discussion Questions

1. How can social media analytics be used in the **consumer products industry**?
2. What do you think are the key challenges, potential solutions, and probable results in applying **social media analytics** in consumer products and services firms?

Social Analytics Social Network Analysis Metrics

- **Connections**

- Homophily
- Multiplexity
- Mutuality/reciprocity
- Network closure
- Propinquity

- **Segmentation**

- Cliques and social circles
- Clustering coefficient
- Cohesion

- **Distribution**

- Bridge
- Centrality
- Density
- Distance
- Structural holes

Social Media Definitions and Concepts

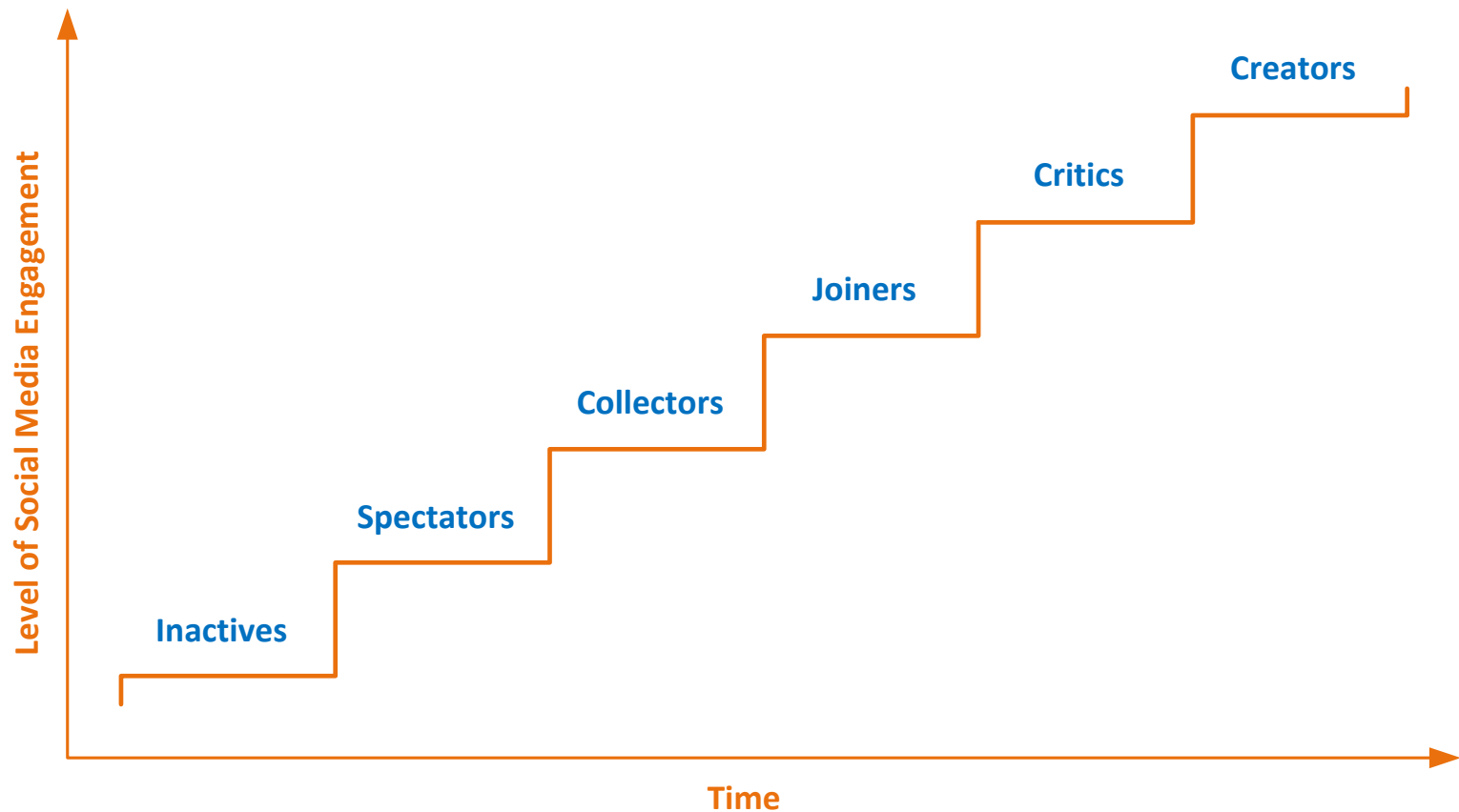
- Enabling technologies of social interactions among people
- Relies on enabling technologies of Web 2.0
- Takes on *many different forms*
 - Internet forums, Web logs, social blogs, microblogging, wikis, social networks, podcasts, pictures, video, and product reviews
- Different types of social media
 - Based on **media research** and **social process**

Social Versus Industrial Media

- Web-based social media are different from **traditional/industrial media**, such as newspapers, television, and film
- Differentiating characteristics
 - Quality
 - Reach
 - Frequency
 - Accessibility
 - Usability
 - Immediacy
 - Updatability

How Do People Use Social Media?

- Different engagement levels



Social Media Analytics

- It is the systematic and scientific ways to consume the vast amount of content created by Web-based social media outlets, tools, and techniques for the betterment of an organization's competitiveness
- Tools to measure social media impact:
 - Descriptive analytics
 - Social network analysis
 - Advanced analytics

Best Practices in Social Media Analytics

- Think of measurement as a guidance system, not a rating system
- Track the elusive sentiment
- Continuously improve the accuracy of text analysis
- Look at the **ripple effect** (spreading results of action)
- Look beyond the brand
- Identify your most powerful **influencers**
- Look closely at the **accuracy** of your analytic tool
- Incorporate **social media intelligence** into planning